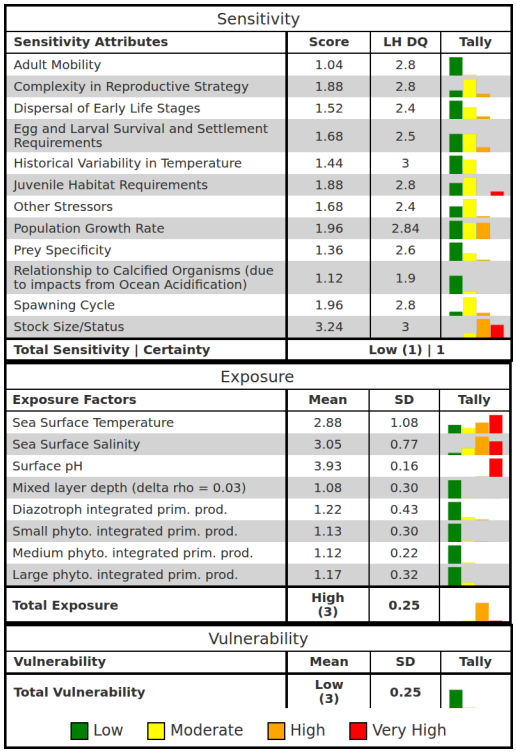
Atlantic mackerel (*Scomber scombrus*)

Summary Scores

- **Sensitivity:** Low
- **Exposure:** High
- **Vulnerability:** Low



Sensitivity

Sensitivity Attributes	Score	LH DQ	Tally
Adult Mobility	1.04	2.8	
Complexity in Reproductive Strategy	1.88	2.8	
Dispersal of Early Life Stages	1.52	2.4	
Egg and Larval Survival and Settlement Requirements	1.68	2.5	
Historical Variability in Temperature	1.44	3	
Juvenile Habitat Requirements	1.88	2.8	
Other Stressors	1.68	2.4	
Population Growth Rate	1.96	2.84	
Prey Specificity	1.36	2.6	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	1.12	1.9	
Spawning Cycle	1.96	2.8	
Stock Size/Status	3.24	3	
Total Sensitivity | Certainty	**Low (1) | 1**		

Exposure

Exposure Factors	Mean	SD	Tally
Sea Surface Temperature	2.88	1.08	
Sea Surface Salinity	3.05	0.77	
Surface pH	3.93	0.16	
Mixed layer depth (delta rho = 0.03)	1.08	0.30	
Diazotroph integrated prim. prod.	1.22	0.43	
Small phyto. integrated prim. prod.	1.13	0.30	
Medium phyto. integrated prim. prod.	1.12	0.22	
Large phyto. integrated prim. prod.	1.17	0.32	
Total Exposure	**High (3)**	**0.25**	

Vulnerability

Vulnerability	Mean	SD	Tally
Total Vulnerability	**Low (3)**	**0.25**	

Figure 1: Summary Sensitivity, Exposure, and Vulnerability Scores and Score Distributions for Atlantic mackerel. Exposure and Vulnerability scores are averaged spatially across the species distribution and within stocks if applicable (see Figure 5 and Figure 4). Life History Data Quality (LH DQ) and Standard Deviations (SD) are reported for senstivity and exposure/vulnerability scores, respectively.

Changes from Previous Assessment(s)

Table 1: Changes to Vulnerability, Exposure, and Sensitivity scores relative to Previous Assessments. If a citation for the previous assessment is not provided, it comes from Hare et al. (2016).

Metric	Previous Assessment(s)	Current Assessment	Change
Sensitivity	Moderate	Low	Decrease
Exposure	High	High	None
Vulnerability	Moderate	Low	Decrease
Directional Effect	Negative	Negative	None

Stocks

Atlantic mackerel is managed as a single stock or stock information was not considered for this iteration of the CVA. ## Score Descriptions:

Sensitivity:

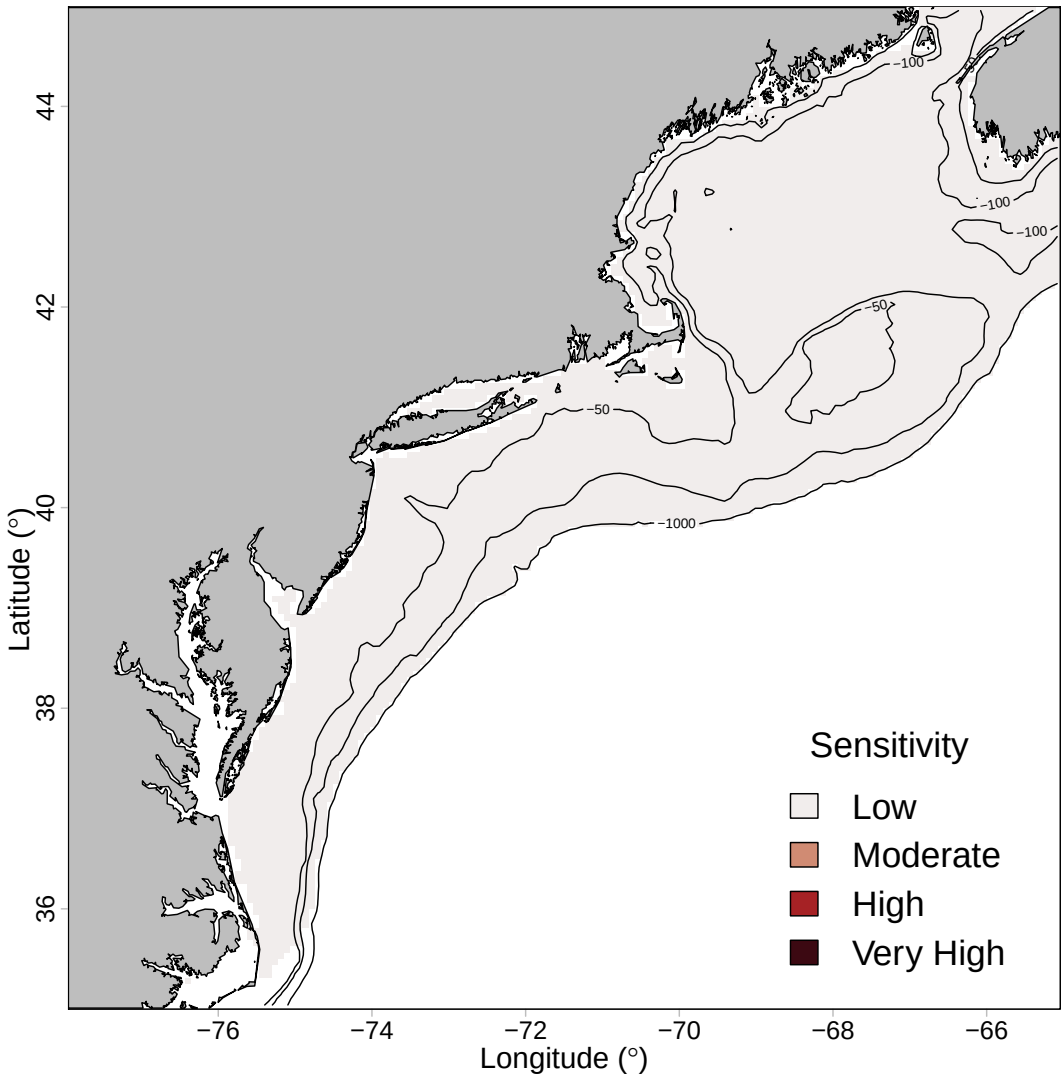


Figure 2: Atlantic mackerel Sensitivity on the Northeast Shelf

Atlantic mackerel received a **Low** sensitivity score with a certainty of 100% (Figure 1). The only score greater than 2.5 was Stock Size (3.2) (Figure 1). Stocks are not currently overfished or experiencing overfishing according to the most recent stock assessment. The stock is rebuilding to target levels (Northeast Fisheries Science Center 2025).

In the previous Climate Vulnerability Assessment, mackerel were given a Moderate sensitivity score (Hare et al. 2016). This previous score was driven by Stock Size, Egg and Larval Survival and Settlement Requirements (Early Life History Requirements), and Spawning Cycle. Stock sizes were not classified at the time of the last assessment, and it was unclear how spawning cycle and early life stages would be impacted by warming waters (Hare et al. 2016). Recent work has suggested that spawning habitats will likely shift west and north under future warming conditions (Bruge et al. 2016).

Exposure:

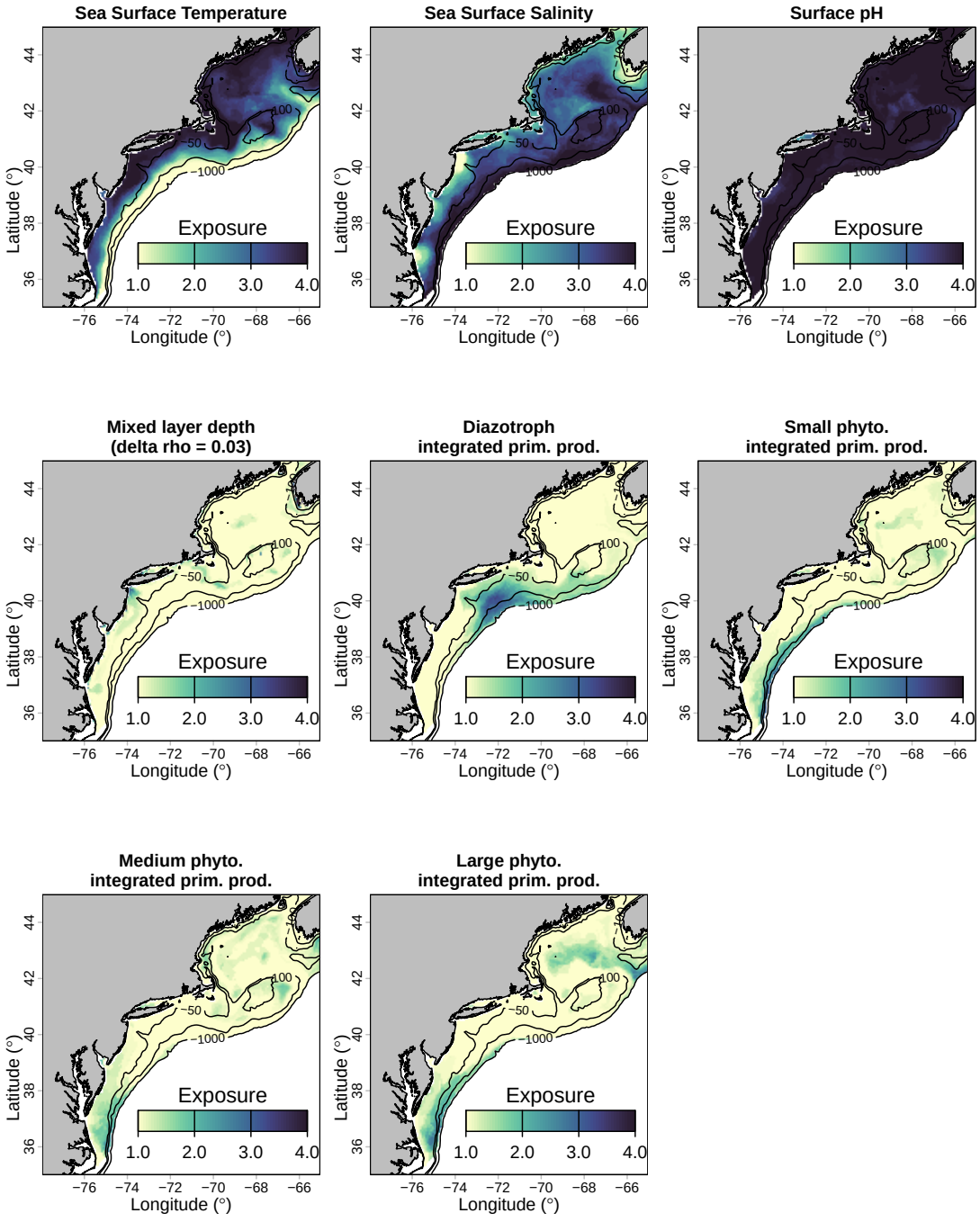


Figure 3: Maps of each Exposure Factor for Atlantic mackerel. Exposure was calculated between 2014-2023 and 2025-2035

Atlantic mackerel had an average Exposure score of 3 (+/- 0.25), resulting in a High Exposure score (Figure 1). Scores were consistent across the shelf, with the highest exposure observed on Georges

Bank (Figure 4). These scores were driven by sea surface temperature, salinity, and pH exposure scores (Figure 3).

In the previous Assessment, mackerel also received a High exposure score, driven by high exposure to changes in sea surface temperatures and pH (Hare et al. 2016).

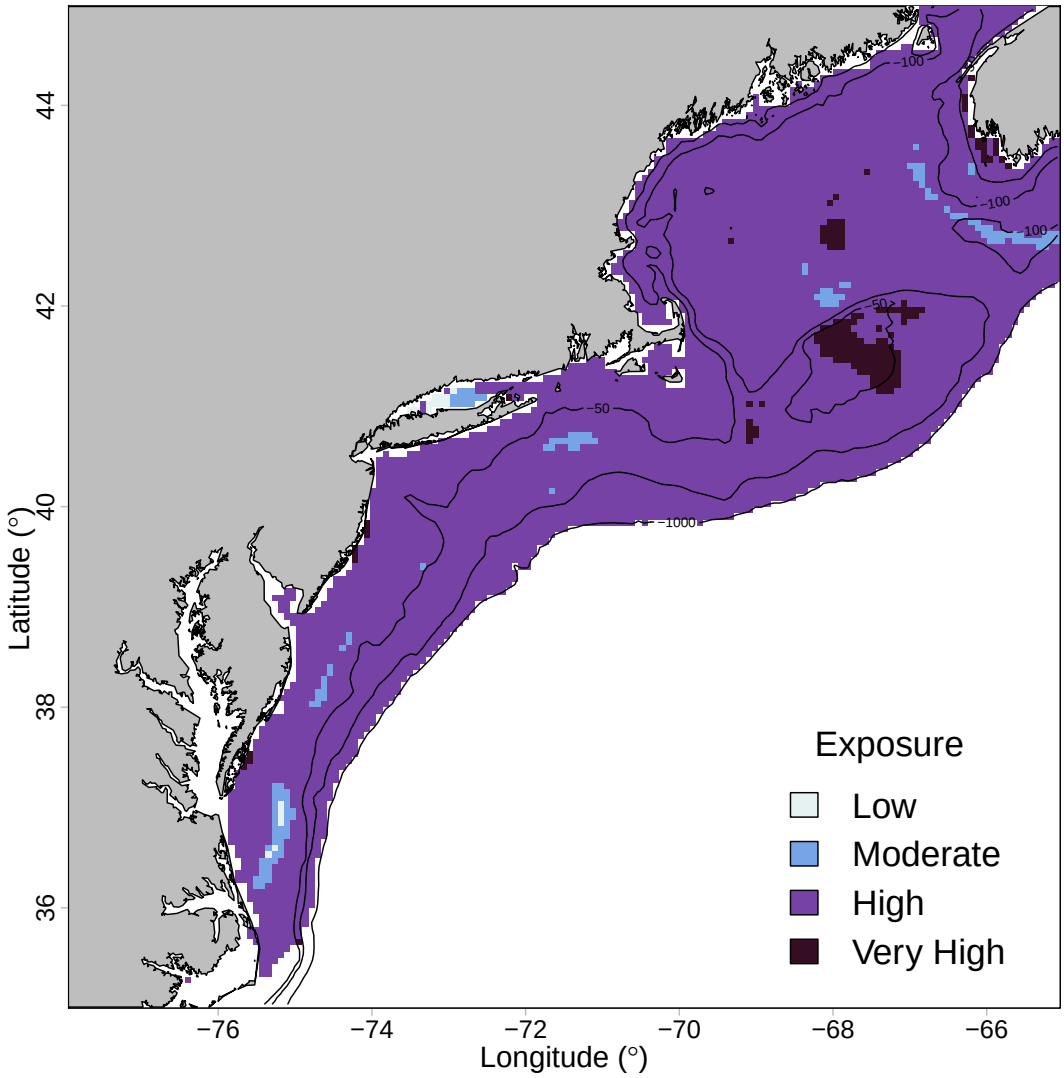


Figure 4: Total Exposure on the Northeast Shelf for Atlantic mackerel. Total Exposure is calculated by applying the logic rule from Hare et al. (2016) within grid cells across exposure factors (see Figure 3)

Vulnerability:

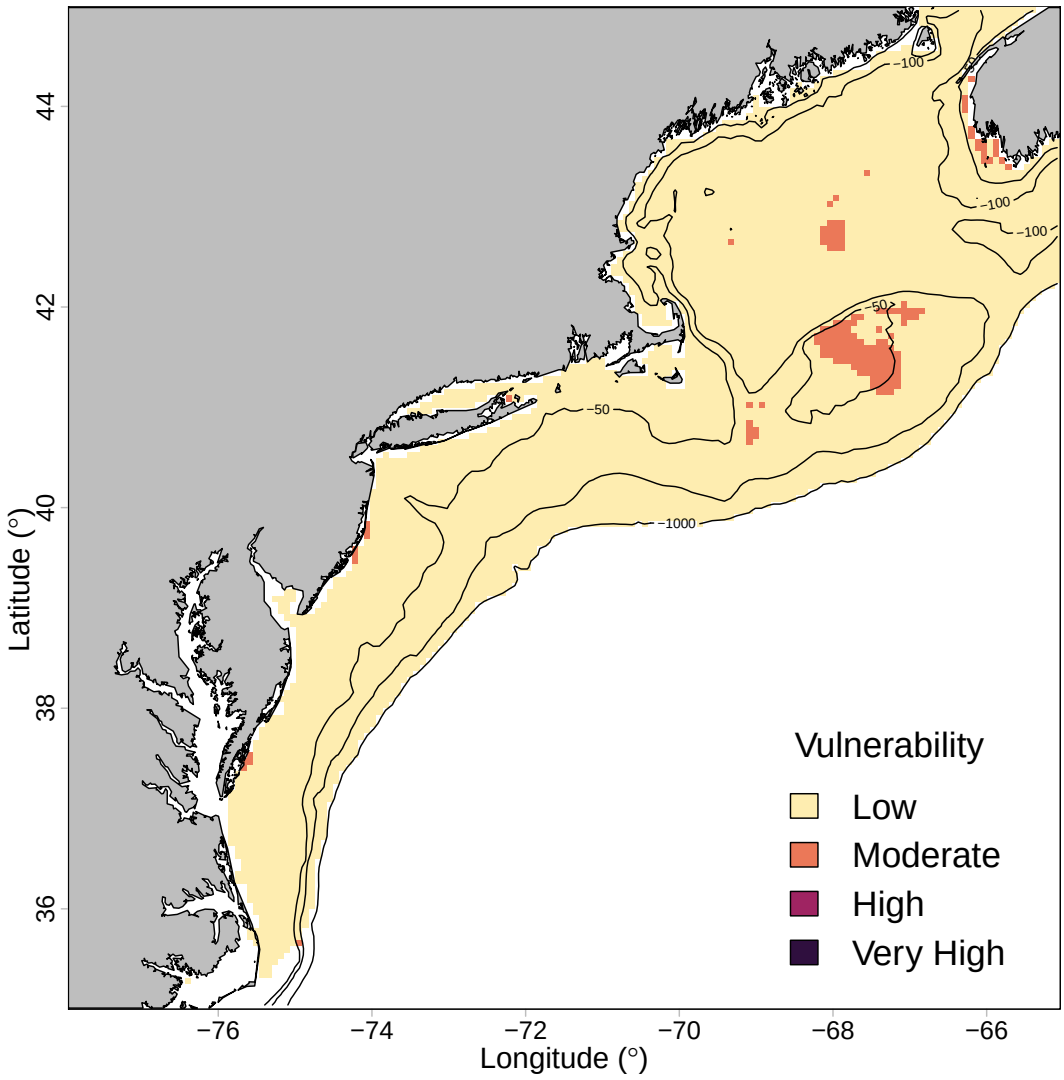


Figure 5: Vulnerability on the Northeast Shelf for Atlantic mackerel. Vulnerability is the product of sensitivity and exposure (see Figure 4 and Figure 2)

The mean overall Vulnerability score was **Low** (3 +/- 0.25) (Figure 1). Scores were low throughout the continental shelf with the exception of Georges Bank, where scores were moderate on portions of the Bank (Figure 5).

In the previous Assessment, mackerel received a Moderate vulnerability score, driven in part by a moderate sensitivity score (Hare et al. 2016).

Data Quality Metrics

Life History Data Quality:

95% of sensitivity attribute data quality scores were greater than or equal to 2, indicating excellent data quality. Only Relationship to Calcified Organisms was considered a data gap (mean attribute data quality score < 2).

Model Confidence Score:

The Atlantic mackerel ensemble species distribution model received a mean model confidence score of 1.4 (+/- 0.5), indicating low confidence in the model. Experts noted that the model missed known seasonal shifts in distribution, but also highlighted that mackerel distributions are patchy and diffuse. The ensemble model's AUC was 0.73, indicating fair performance.

Additional Metrics

Distribution Shifts:

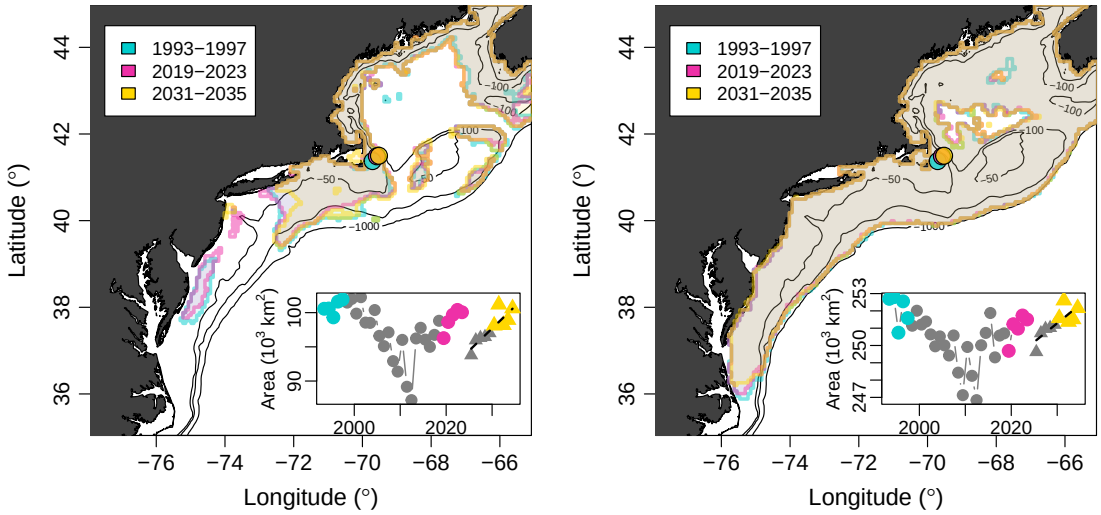


Figure 6: Distribution shifts on the Northeast Shelf for Atlantic mackerel. Polygons illustrate core habitat (left) and species range (right) for five-year average periods. Points illustrate weighted centroid, calculated as the average position with species distribution models as weights. Inset timeseries illustrate the shift in core and habitat range area over time. Circles represent points from the MOM6 hindcast; triangles represent points from the MOM6 forecast. Dashed lines represent significant linear regressions

Note: For an animated version of this plot, please see the online version of this document.

Mackerel habitats have been highly variable between 1993 and 2023. Habitat areas declined around 2010 and have since recovered to near 1993 levels. The southern edge of mackerel ranges have moved northward (Figure 6). The weighted centroid shifted 16 km northeast between 1993 and 2023.

Mackerel habitats are predicted to expand in future ocean conditions. Both core habitat area and mackerel range are predicted to increase. While the southern edge of the range is predicted to continue to move northward, mackerel are expected to move into new areas in southern New England (Figure 6). Distributions are predicted to continue to shift eastward by 5 km.

In the previous Assessment, mackerel received a High Distributional Vulnerability rank, due to their high mobility, migrations, and high larval dispersal (Hare et al. 2016).

Table 2: Distribution shifts based on five-year average distributions. Changes are calculated relative to the previous time period.

Time Period	Weighted Centroid Shift (km & bearing)	Core Habitat Area (change) (km^2)	Habitat Total Area (change) (km^2)	Latitude Range (change) (deg)	Change in Longitude Range (deg)
1993-1997	NA	101,852	252,291	6.80	9.06
2019-2023	16.1 km NE	99,283 (-2,569)	251,261 (-1,030)	6.61 (-0.19)	9.14 (+0.08)
2031-2035	5.3 km ENE	99,418 (+135)	251,856 (+594)	6.55 (-0.06)	9.06 (-0.08)

Directional Effect:

Atlantic mackerel are likely to be **negatively** impacted by climate change. Expert certainty was 83%. Regional warming has resulted in northeasterly shifts in mackerel distributions. Spawning habitat could increase in more northern regions such as the Gulf of St Lawrence as mackerel seek out preferred temperatures. This result is similar to the results from the 2016 Climate Vulnerability Assessment for the region (Hare et al. 2016).

Climate Effects on Abundance and Distribution

Atlantic mackerel are sensitive to temperature changes, particularly temperatures greater than 5°C (Overholtz et al. 2011). From 1968 to 2008, the distribution of mackerel in the Northwest Atlantic has shifted roughly 250km north and east in addition to changes in depth distribution to more shallow on-shelf waters. These changes in position and depth are known to be correlated with the gradual warming of the Northwest Atlantic (Overholtz et al. 2011). Atlantic mackerel egg distribution has also shifted northeastward as a consequence of warming, depleting egg production in important spawning locations in southern New England (Richardson et al. 2020). Bruge et al. (2016) projected changes in Northeast Atlantic mackerel spawning distribution under future climate scenarios, and found that the spawning center of gravity is expected to shift westward and northward. Mbaye et al. (2019) investigated effects of environmental conditions on Atlantic mackerel spawning habitat in the Gulf of St. Lawrence. Generalized additive model (GAM) results showed that spawning habitat in the Gulf of St. Lawrence could expand and extend to western and southern Newfoundland.

Life History Synopsis

Atlantic mackerel in the Northwest Atlantic inhabit cold and temperate continental shelf waters from the Gulf of St Lawrence to Cape Lookout, North Carolina (Studholme 1999). Spawning occurs from April to August in the Northwest Atlantic and progresses from south to north with warming temperatures and following migration patterns (Sette 1943). There are two spawning contingents: the southern contingent

spawns from April to July in the Mid-Atlantic Bight and Gulf of Maine, and the northern contingent spawns in the southern Gulf of St. Lawrence in June and July (Berrien 1982) (Morse et al. 1987). Temperature is a cue for spawning. Spawning does not occur until water has warmed to 8°C with most egg production occurring between 9-14°C (Sette 1943). More recent work has shown that mackerel spawning and migration timing is negatively correlated with sea surface temperature (Jansen and Gislason 2011). Early life stage development happens quickly, with eggs hatching in 4-7.5 hours, depending on temperature (Berrien 1982). Larvae (< 13 mm) occur primarily in offshore waters from Chesapeake Bay to southern Gulf of St. Lawrence (Sette 1943). Larval and juvenile depth varies diurnally and with age (Collette and Klein-MacPhee 2002). Larvae feed primarily on zooplankton and copepods, and larvae > 6.4 mm are known to experience cannibalistic behavior (Castonguay et al. 2008). Juveniles rely on water column habitats, both on the shelf, and in estuaries (Guyant et al. 2022). Juveniles and adults are opportunistic feeders that prey on a wide range of crustaceans, copepods, planktonic organisms, and fishes (Sette 1950). Squid, seabirds and larger piscivores, including Atlantic cod, prey on juvenile mackerel (Studholme 1999). Adults over winter in moderately deep water, 70-200 m, along the continental shelf. After migrating inshore in April-May, they form the two spawning aggregations (Berrien 1982). Predators of adult mackerel include larger fish, sharks, skates, seabirds, and marine mammals (primarily whales and seals) (Studholme 1999). Atlantic mackerel are managed by the Mid-Atlantic Fishery Management Council under the Mackerel, Squid, and Butterfish Fishery Management Plan. As of the 2025 assessment, the stock is not overfished and overfishing is not occurring, but is rebuilding to target levels (Northeast Fisheries Science Center 2025).

Supplemental Scores

Exposure Timeseries

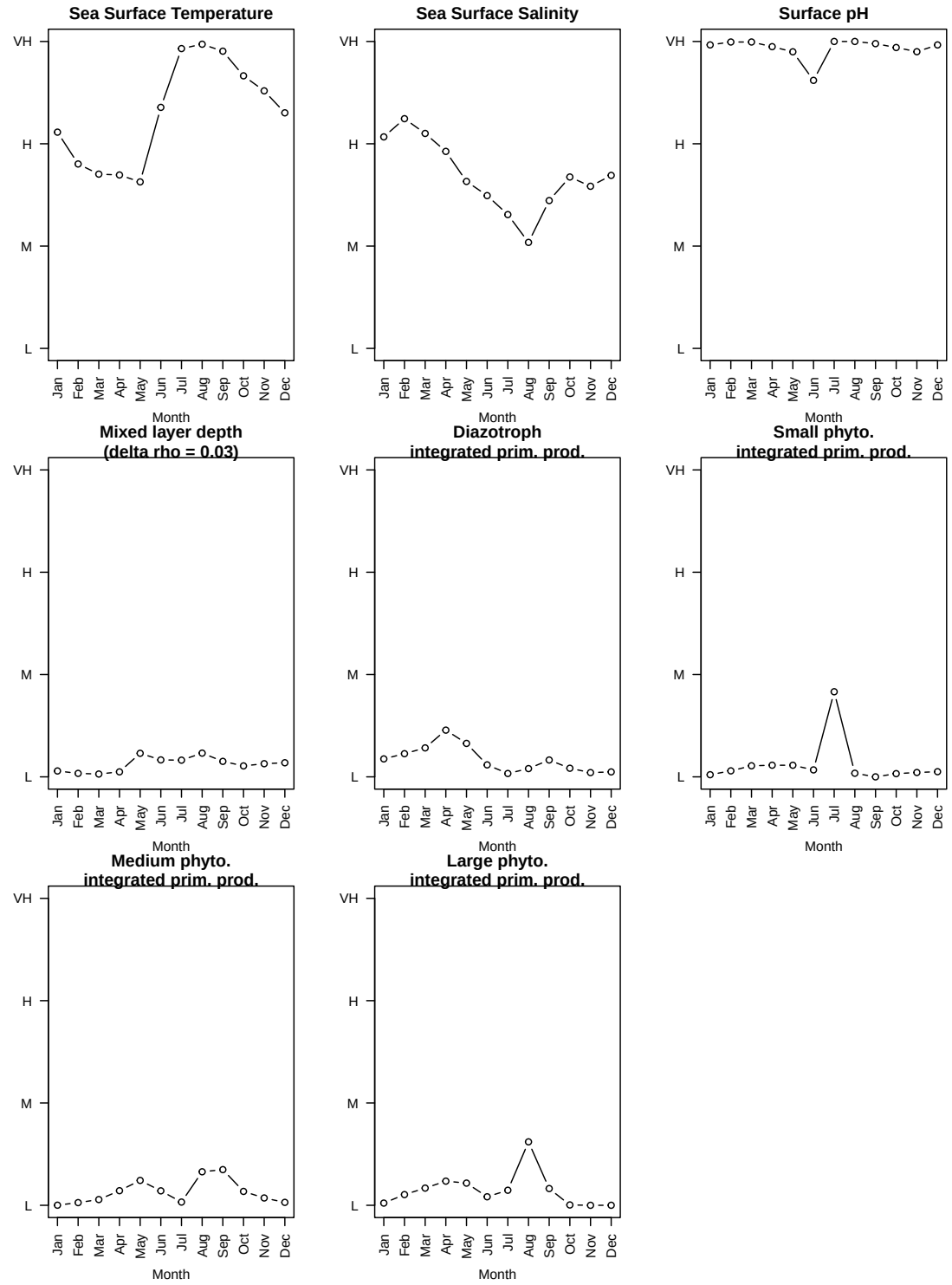


Figure 7: Monthly Timeseries of each Exposure Factor for Atlantic mackerel. Exposure was calculated between 2014-2023 and 2025-2035

Monthly exposure was consistent for Atlantic mackerel. Exposure was High most of the year, with the exception of April and May, where exposure declined to Moderate due to declines in sea surface temperature exposure (Figure 8, Figure 7).

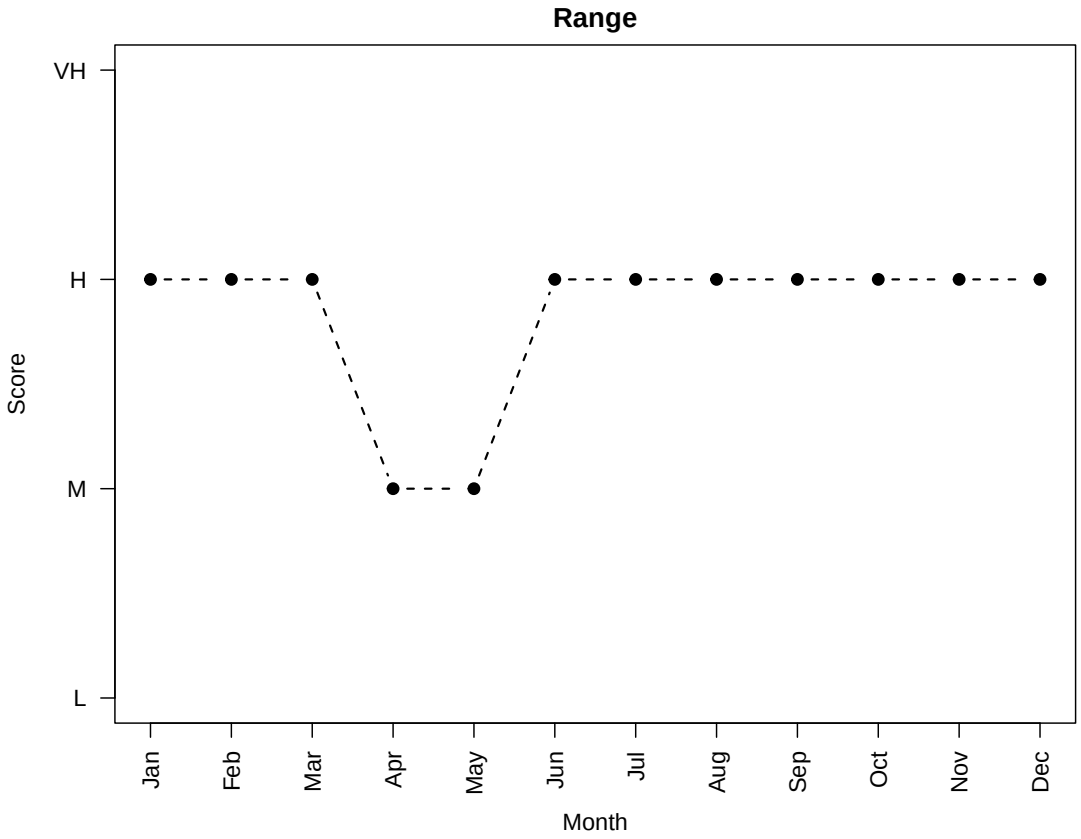


Figure 8: Timeseries of Total Exposure for Atlantic mackerel

Exposure due to Significant Factors

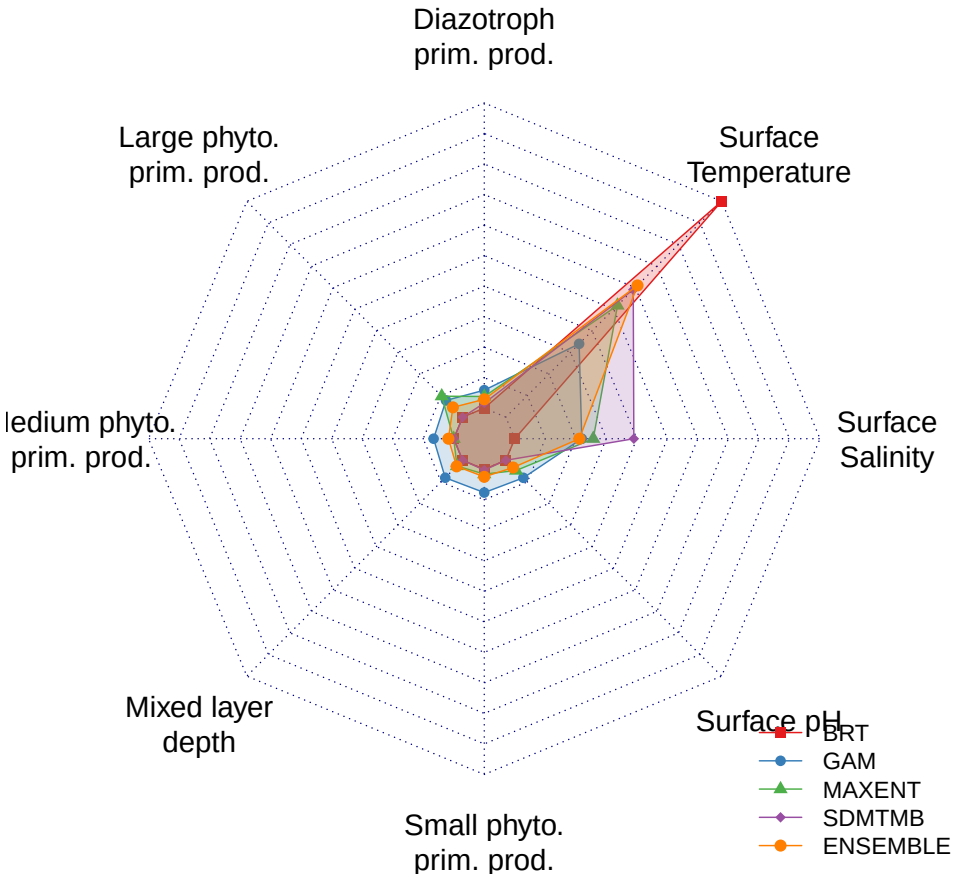


Figure 9: Dynamic Factors driving Atlantic mackerel distributions. The closer the polygon is to the edge of the figure, the more important the variable is. Each ring represents 10% variance explained. Factors must explain at least 10% of variance in the ensemble model to be used to calculate total exposure.

Sea surface temperatures and salinities were the major drivers of mackerel distributions (Figure 9). Mackerel are sensitive to sea surface temperatures (Overholtz et al. 2011; Jansen and Gislason 2011), and juveniles rely on estuarine habitats where they will be exposed to changes in salinity (Guyant et al. 2022). When exposure was calculated with only these factors, the mean exposure score decreased to Moderate (1.71 +/- 0.85) (Figure 11). Exposure scores were greatest in the Gulf of Maine and on the mid continental shelf where high exposure to temperature and salinity changes overlapped (Figure 10).

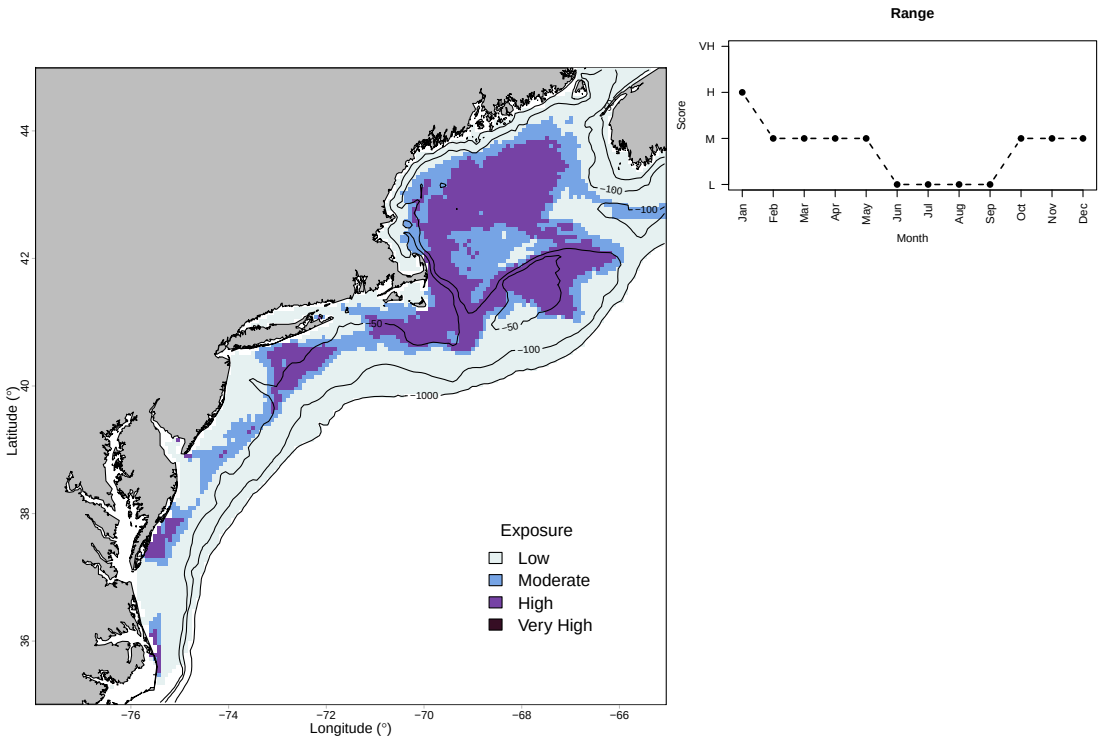


Figure 10: Total Exposure Map and Timeseries when Exposure was calculated with only Exposure Factors critical to Atlantic mackerel distributions

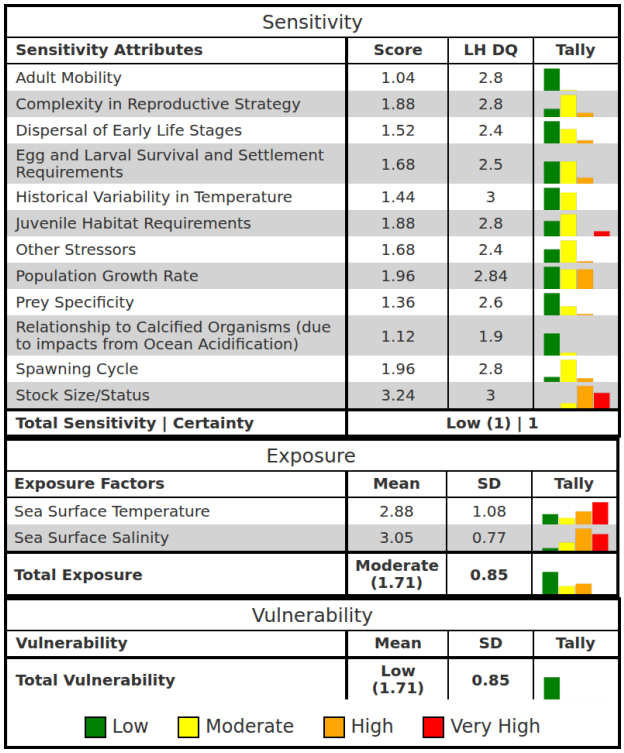


Figure 11: As in Figure 1 for Atlantic mackerel, but with only significant Exposure factors and the resulting Vulnerability Scores.

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